ISA BUS

BUS1
PCB Edge Connector

8-BIT ISA

1	GND	IO_CH_CK	32
2	RESETDRV	SD7	33
3	GND	SD6	34
4	VCC	SD5	35
5	IRQ2	SD4	36
6	-5V	SD3	37
7	DRQ2	SD2	38
8	VSS	SD1	39
9	-12V	SD0	40
10	VDD	IO_READY	41
11	GND	AEN	42
12	MEMW	SA19	43
13	MEMR	SA18	44
14	IOW	SA17	45
15	IOR	SA16	46
16	DACK3	SA15	47
17	DRQ3	SA14	48
18	DACK1	SA13	49
19	DRQ1	SA12	50
20	DACK0	SA11	51
21	CLK	SA10	52
22	IRQ7	SA9	53
23	IRQ6	SA8	54
24	IRQ5	SA7	55
25	IRQ4	SA6	56
26	IRQ3	SA5	57
27	DACK2	SA4	58
28	TC	SA3	59
29	ALE	SA2	60
30	VCC	SA1	61
31	OSC	SA0	62
32	GND		

C27 10uF 10% 50VDC X7R-MLCC LS: 5mm
C28 10uF 10% 50VDC X7R-MLCC LS: 5mm

FDC CHIP SELECT

C5 0.1uF 10% 50VDC X7R-MLCC LS: 5mm
C4 0.1uF 10% 50VDC X7R-MLCC LS: 5mm

U5 74ALS138
U4 74ALS138

A8 A4 A3 A5 AEN A6 A7 A9 GND VCC

FDC_CS

FLOPPY DISK CONTROLLER

U1 82077AA

X1	33	INVERT	35
X2	34		
RESETDRV	32	ME0	57
		ME1	61
D0	11	ME2	63
D1	13	ME3	66
D2	14		
D3	15	DS0	58
D4	17	DS1	62
D5	19	DS2	64
D6	20	DS3	67
D7	22		
		HDSEL	51
A0	7	STEP	55
A1	8	DIR	56
A2	10	WRDATA	53
		WE	52
FDC_CS	6	DENSEL	49
IOR	4	DSKCHG	31
IOW	5	DRV2	30
		TRK0	2
DRQ2	24	WP	1
DACK2	3	INDX	26
TC	25		
IRQ6	23	RDDATA	41
		HIFIL	38
27		LOFIL	37
		MFM	48
IDENT		DRATE0	28
		DRATE1	29
AVCC	46	PLL0	39
AVSS	45		

C16 22pF 10% 50VDC X7R-MLCC LS: 5mm
C15 22pF 10% 50VDC X7R-MLCC LS: 5mm
X1 24MHz
C1 0.1uF 10% 50VDC X7R-MLCC LS: 5mm
C17 4.7nF 10% 50VDC X7R-MLCC LS: 5mm
RR1 1k

JP1 PS/2 FLOPPY 1 2 FLOPPY_VCC

P1 FLOPPY

DENSEL	2	1	
	4	3	
	6	5	
INDEX	8	7	
MEA	10	9	
DSB	12	11	
DSA	14	13	
MEB	16	15	
DIR	18	17	
STEP	20	19	
WDATA	22	21	
WE	24	23	
TRK00	26	25	
WP	28	27	
RDATA	30	29	
HDSEL	32	31	
DSKCG	34	33	

XT-CF CHIP SELECT

U9A 74LS04
U10B 74LS32
U10A 74LS32
U8 74LS688

A4 A5 A6 A7 A8 A9 GND VCC AEN

CF_CS1 CF_CS0

RR3 10k
SW2 XTCF_CS

SW2.1 – ON: CF A7 = 0: OFF: CF A7 = 1
SW2.2 – ON: CF A6 = 0: OFF: CF A6 = 1
SW2.3 – ON: CF A5 = 0: OFF: CF A5 = 1

Spares

U10C 74LS32
U10D 74LS04
U9B 74LS04

C8 0.1uF 10% 50VDC X7R-MLCC LS: 5mm
C10 0.1uF 10% 50VDC X7R-MLCC LS: 5mm

FDC BIOS ROM

RR2 10kΩ 2% 1/8W SIP-10 Bussed (9 Resistor) LS: 2.54mm
C3 0.1uF 10% 50VDC X7R-MLCC LS: 5mm
C2 0.1uF 10% 50VDC X7R-MLCC LS: 5mm

U3 74LS688
U2 28C64

A19 A18 A17 A16 A15 A14 A13 A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12

ROM_CS_IN
FDC_RDY/BUSY
FDC_ROM_CS
FDC_ROMW

SW1
ROM_CS_IN MEMW FDC_CFG

SW1.1 – ON: Enable ROM
SW1.2 – ON: Enable EEPROM write
SW1.3 – ON: ROM A17 = 0: OFF: ROM A17 = 1
SW1.4 – ON: ROM A16 = 0: OFF: ROM A16 = 1
SW1.5 – ON: ROM A15 = 0: OFF: ROM A15 = 1
SW1.6 – ON: ROM A14 = 0: OFF: ROM A14 = 1
SW1.7 – ON: ROM A13 = 0: OFF: ROM A13 = 1
SW1.8 – NOT USED

XT-CF BIOS ROM

RR4 10kΩ 2% 1/8W SIP-10 Bussed (9 Resistor) LS: 2.54mm
C7 0.1uF 10% 50VDC X7R-MLCC LS: 5mm
C6 0.1uF 10% 50VDC X7R-MLCC LS: 5mm

U7 74LS688
U6 28C64

CF_RDY/BUSY
CF_ROMW
CF_ROM_CS

SW3
MEMW XTCF_CFG

SW3.1 – ON: Enable ROM
SW3.2 – ON: Enable EEPROM write
SW3.3 – ON: ROM A17 = 0: OFF: ROM A17 = 1
SW3.4 – ON: ROM A16 = 0: OFF: ROM A16 = 1
SW3.5 – ON: ROM A15 = 0: OFF: ROM A15 = 1
SW3.6 – ON: ROM A14 = 0: OFF: ROM A14 = 1
SW3.7 – ON: ROM A13 = 0: OFF: ROM A13 = 1
SW3.8 – NOT USED

CF INTERFACE

C18 0.1uF 10% 50VDC X7R-MLCC LS: 5mm

P2 CF_CARD

A1	20	D0	21
A2	19	D1	22
A3	18	D2	23
GND	17	D3	2
GND	16	D4	3
GND	15	D5	4
GND	14	D6	5
GND	12	D7	6
GND	11	D8	47
GND	10	D9	48
GND	8	D10	49
		D11	27
CF_CS0	7	D12	28
CF_CS1	32	D13	29
		D14	30
		D15	31
GND	9	DASP	45
GND	39	PDIAG	46
		CD1	26
VCC	44	CD2	25
		VS1	33
VCC	36	VS2	40
IOR	34	IOCS16	24
IOW	35		
		INTRQ	37
CF_RESET	41	IORDY	42
		INPACK	43

CF_DASP CF_IORDY CF_DMARQ

R3 5.6kΩ 1% 1/4W Axial – Metal Film 6.3mm
R5 1kΩ 1% 1/4W Axial – Metal Film 6.3mm
R4 10kΩ 1% 1/4W Axial – Metal Film 6.3mm
R2 330Ω 1% 1/4W Axial – Metal Film 6.3mm

U9C 74LS04
U9D 74LS04
U9E 74LS04
U9F 74LS04

RESETDRV CF_RESET CF_DASP

C9 0.1uF 10% 50VDC X7R-MLCC LS: 5mm

JP2 LED_XTCF

XT-USB

SW5
IRQ3 IRQ4 IRQ5 IRQ6 IRQ7 USB_IRQ

SW5.1 – ON: IRQ3
SW5.2 – ON: IRQ4
SW5.3 – ON: IRQ5
SW5.4 – ON: IRQ6
SW5.5 – ON: IRQ7
Only one switch can be ON at a time
If everything OFF: no IRQ

R6 1kΩ 1% 1/4W Axial – Metal Film 6.3mm
R7 1kΩ 1% 1/4W Axial – Metal Film 6.3mm
Q1 2N7000
74LS32 U14A

U11 CH375

USB-CS IOR IOW A0 USB-D0 USB-D1 USB-D2 USB-D3 USB-D4 USB-D5 USB-D6 USB-D7 RESETDRV ACT

C11 0.1uF 10% 50VDC X7R-MLCC LS: 5mm
C21 0.01uF 10% 50VDC X7R-MLCC LS: 5mm
C29 0.1uF 10% 50VDC X7R-MLCC LS: 5mm
C25 22pF 10% 50VDC X7R-MLCC LS: 5mm
C26 22pF 10% 50VDC X7R-MLCC LS: 5mm
X2 12MHz
C20 22pF 10% 50VDC X7R-MLCC LS: 5mm
C19 22pF 10% 50VDC X7R-MLCC LS: 5mm

P3 USB_A
U15 USBLC6-2P6
C24 0.1uF 10% 50VDC X7R-MLCC LS: 5mm
C23 10uF
FB1 FILTER
F1 1.1A 16VDC PPTC Resettable Fuses LS: 5.8mm
C22 10uF

C12 0.1uF 10% 50VDC X7R-MLCC LS: 5mm
U12 74HC245

USB-D0 USB-D1 USB-D2 USB-D3 USB-D4 USB-D5 USB-D6 USB-D7 IOR USB-CS

C13 0.1uF 10% 50VDC X7R-MLCC LS: 5mm
U13 74ALS138

SW4
SW4.1 – ON: 230h
SW4.2 – ON: 240h
SW4.3 – ON: 250h
SW4.4 – ON: 260h
Only one switch can be ON at a time
USB_CFG

U14B 74LS32
U14C 74LS32
U14D 74LS32

A3 AEN USB-CS

C14 0.1uF 10% 50VDC X7R-MLCC LS: 5mm

R8 470Ω 1% 1/4W Axial – Metal Film 6.3mm
ACT
JP3 LED_USB

R1 470Ω 1% 1/4W Axial – Metal Film 6.3mm
PWR_LED
D1

Mount Holes

HOLE1
HOLE2

Design by:	
FDC: Sergey Kiselev	
XT-CF lite V4.0c: Sergey Kiselev	
Merged and added CH375B USB-Storage: Peter Reichenbach	
Sheet: /	
File: isa_FDC-CF-USB.kicad_sch	
Title: ISA_FDC-CF-USB Card	**V1.0**
Size: A2	Date: 2026 October 17
KiCad E.D.A. 10.0.7	Rev: A Id: 1/1